

1. Grade Checker

Take a score as input and print the grade based on the following:

90+ : "A"

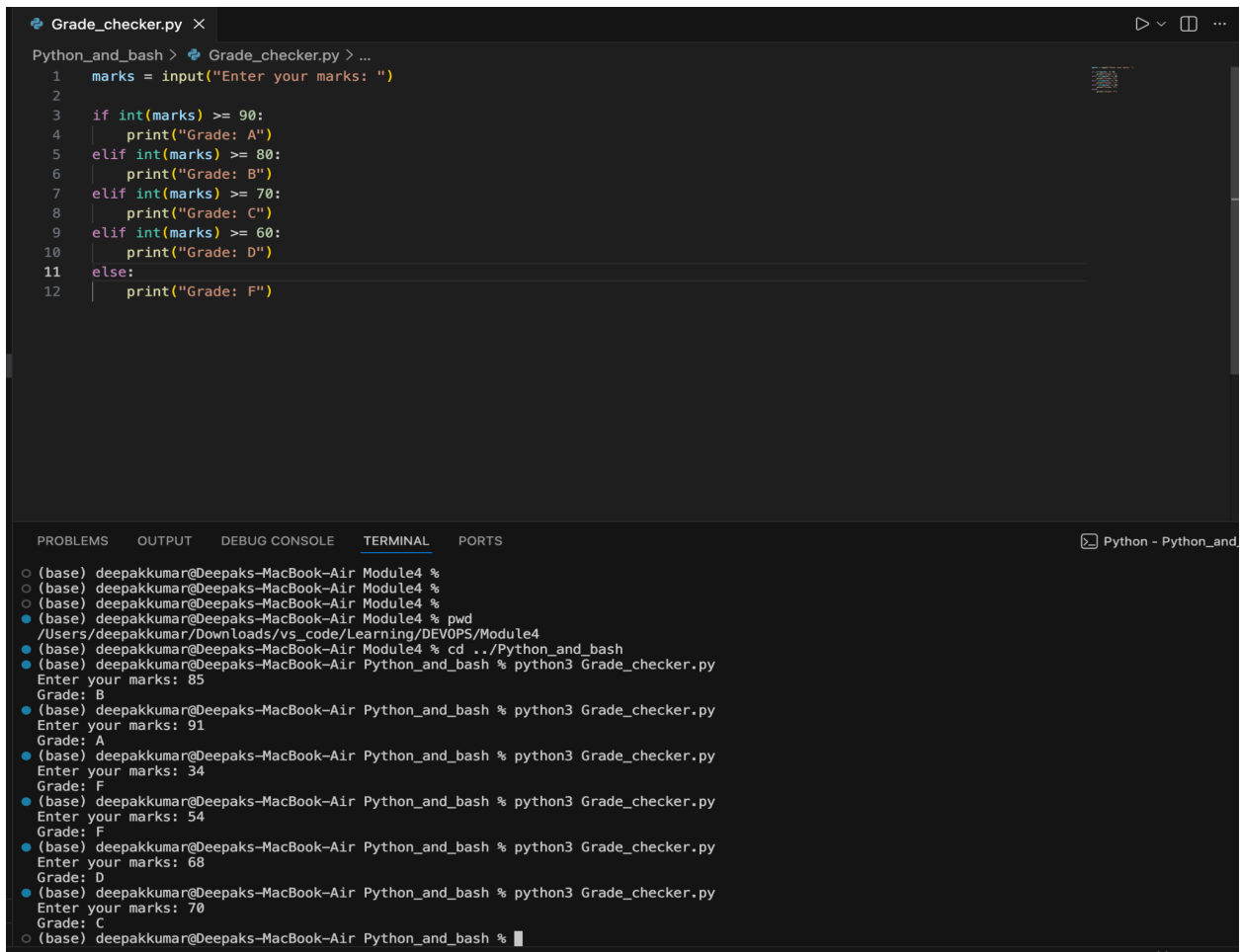
80-89 : "B"

70-79 : "C"

60-69 : "D"

Below 60 : "F"

here we used a basic if else statement to carry out marks and all.



```
Grade_checker.py x
Python_and_bash > Grade_checker.py > ...
1 marks = input("Enter your marks: ")
2
3 if int(marks) >= 90:
4     print("Grade: A")
5 elif int(marks) >= 80:
6     print("Grade: B")
7 elif int(marks) >= 70:
8     print("Grade: C")
9 elif int(marks) >= 60:
10    print("Grade: D")
11 else:
12    print("Grade: F")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python - Python_and_
o (base) deepakkumar@Deepaks-MacBook-Air Module4 %
o (base) deepakkumar@Deepaks-MacBook-Air Module4 %
o (base) deepakkumar@Deepaks-MacBook-Air Module4 %
o (base) deepakkumar@Deepaks-MacBook-Air Module4 % pwd
/Users/deepakkumar/Downloads/vs_code/Learning/DEVOPS/Module4
o (base) deepakkumar@Deepaks-MacBook-Air Module4 % cd ../Python_and_bash
o (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % python3 Grade_checker.py
Enter your marks: 85
Grade: B
o (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % python3 Grade_checker.py
Enter your marks: 91
Grade: A
o (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % python3 Grade_checker.py
Enter your marks: 34
Grade: F
o (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % python3 Grade_checker.py
Enter your marks: 54
Grade: F
o (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % python3 Grade_checker.py
Enter your marks: 68
Grade: D
o (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % python3 Grade_checker.py
Enter your marks: 70
Grade: C
o (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash %
```

2 Student Grades

Create a dictionary where the keys are student names and the values are their grades.

Allow the user to:

Add a new student and grade.

Update an existing student's grade.

Print all student grades.

Used dictionary and basic operations. Using if else:

CODE:

```
students={}

while True:

    print("\nStudent Grade Management System"

        "\n1. Add Student"

        "\n2. Update Student Grade"

        "\n3. View Students"

        "\n4. Exit")

    choice = input("Choose an option (1-4): ")

    if choice == '1':

        name = input("Enter student name: ")

        grade = input("Enter student grade: ")
```

```
students[name] = grade

print(f"Student {name} added with grade {grade}.")

elif choice == '2':

    name = input("Enter student name to update: ")

    if name in students:

        grade = input("Enter new grade: ")

        students[name] = grade

        print(f"Student {name}'s grade updated to {grade}.")

    else:

        print(f"Student {name} not found.")

elif choice == '3':

    if students:

        print("\nStudent Grades:")

        for name, grade in students.items():

            print(f"{name}: {grade}")

    else:

        print("No students found.")

elif choice == '4':

    print("Exiting the program.")

    break
```

Output :

Student Grade Management System

1. Add Student

2. Update Student Grade

3. View Students

4. Exit

Choose an option (1-4): 3

No students found.

Student Grade Management System

1. Add Student

2. Update Student Grade

3. View Students

4. Exit

Choose an option (1-4):

Student Grade Management System

1. Add Student

2. Update Student Grade

3. View Students

4. Exit

Choose an option (1-4):

Student Grade Management System

1. Add Student

2. Update Student Grade

3. View Students

4. Exit

Choose an option (1-4): 3

No students found.

Student Grade Management System

1. Add Student

2. Update Student Grade

3. View Students

4. Exit

Choose an option (1-4):

Student Grade Management System

1. Add Student
2. Update Student Grade
3. View Students
4. Exit

Choose an option (1-4):

Student Grade Management System

1. Add Student
2. Update Student Grade
3. View Students
4. Exit

Choose an option (1-4):

Student Grade Management System

1. Add Student
2. Update Student Grade
3. View Students
4. Exit

Choose an option (1-4): 3

No students found.

Student Grade Management System

1. Add Student
2. Update Student Grade
3. View Students
4. Exit

Choose an option (1-4): 1

Enter student name: Ram

Enter student grade: A

Student Ram added with grade A.

Student Grade Management System

1. Add Student
2. Update Student Grade
3. View Students
4. Exit

Choose an option (1-4): 1

Enter student name: Shayam

Enter student grade: B

Student Shayam added with grade B.

Student Grade Management System

1. Add Student
2. Update Student Grade
3. View Students
4. Exit

Choose an option (1-4): 1

Enter student name: Mohan

Enter student grade: C

Student Mohan added with grade C.

Student Grade Management System

1. Add Student
2. Update Student Grade
3. View Students
4. Exit

Choose an option (1-4): 1

Enter student name: Sita

Enter student grade: D

Student Sita added with grade D.

Student Grade Management System

1. Add Student
2. Update Student Grade
3. View Students
4. Exit

Choose an option (1-4): 3

Student Grades:

Ram: A

Shayam: B

Mohan: C

Sita: D

Student Grade Management System

1. Add Student
2. Update Student Grade
3. View Students
4. Exit

Choose an option (1-4): 2

Enter student name to update: Sita

Enter new grade: A

Student Sita's grade updated to A.

Student Grade Management System

1. Add Student

2. Update Student Grade

3. View Students

4. Exit

Choose an option (1-4): 3

Student Grades:

Ram: A

Shayam: B

Mohan: C

Sita: A

Student Grade Management System

1. Add Student

2. Update Student Grade

3. View Students

4. Exit

Choose an option (1-4): 3

Student Grades:

Ram: A

Shayam: B

Mohan: C

Sita: A

Student Grade Management System

1. Add Student

2. Update Student Grade

3. View Students

4. Exit

Choose an option (1-4): 3

Student Grades:

Ram: A

Shayam: B

Mohan: C

Sita: A

Student Grade Management System

1. Add Student

2. Update Student Grade

3. View Students

4. Exit

Choose an option (1-4): 3

Student Grades:

Ram: A

Shayam: B

Mohan: C

Sita: A

Student Grade Management System

1. Add Student

2. Update Student Grade

3. View Students

4. Exit

```
(base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % python3 students_grade.py
```

```
Student Grade Management System
1. Add Student
2. Update Student Grade
3. View Students
4. Exit
Choose an option (1-4): 1
Enter student name: Sita
Enter student grade: D
Student Sita added with grade D.
```

```
Student Grade Management System
1. Add Student
2. Update Student Grade
3. View Students
4. Exit
Choose an option (1-4): 3
```

```
Student Grades:
Ram: A
Shayam: B
Mohan: C
Sita: D
```

```
Student Grade Management System
1. Add Student
2. Update Student Grade
3. View Students
4. Exit
Choose an option (1-4): 2
Enter student name to update: Sita
Enter new grade: A
Student Sita's grade updated to A.
```

```
Student Grade Management System
1. Add Student
2. Update Student Grade
3. View Students
4. Exit
Choose an option (1-4): 3
```

```
Student Grades:
Ram: A
Shayam: B
Mohan: C
```

3. Write to a File

Write a program to create a text file and write some content to it.

Using file functions like write and open.

CODE:

```
import os

if not os.path.exists("sample_output.txt"):

    open("sample_output.txt", "w").close()

data = open("sample_output.txt", "r+")

data.write("This is first line\n")

data.write("This is second line\n")

data.write("This is third line\n")

data.close()

file = open("sample_output.txt", "r")

content = file.read()

print(content)

file.close()
```

Output:

```
○ (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash %  
● (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % python3 write_file.py  
This is first line  
This is second line  
This is third line  
  
○ (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % █
```

4. Read from a File

We used open in read mode and file.read to read and print to display.

```
file = open("sample_output.txt", "r")  
  
content = file.read()  
  
print(content)  
  
file.close()
```

Output:

```
○ (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash %  
● (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % python3 write_file.py  
This is first line  
This is second line  
This is third line  
  
○ (base) deepakkumar@Deepaks-MacBook-Air Python_and_bash % █
```

Submission Guidelines :- Attach Screenshots or command along with explanation and submit in doc(google doc or microsoft doc) format or share github link